

Mauna Loa in Hawaii is an ideal site for an atmospheric research station. The high altitude and remote location of the volcano ensure that the air is largely unaffected by humans or vegetation.

Seasonal changes

Although South Pole funding ended in 1964, Mauna Loa has produced data from 1958 onward. Plotted on a graph, the measurements became known as the Keeling Curve. It is, in fact, a series of annual curves, reflecting seasonal changes. During spring and summer in the Northern Hemisphere, as new foliage takes more CO_2 out of the atmosphere, the global concentration of the gas declines, reaching a low point in September. It increases again in the northern autumn as leaves fall and photosynthesis declines. Plant growth in the Southern Hemisphere later in the year does not make up for the loss, because most of Earth's vegetative cover is in the north.

Ancient air bubbles trapped in polar ice cores reveal that, over the past 11,000 years, average CO_2 concentrations were 275–285 ppmv, but